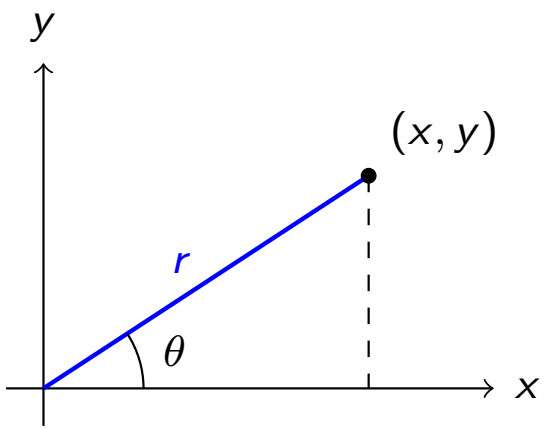
The Laplacian in Polar Coordinates

- For circular regions we put $x = r\cos\theta$, $y = r\sin\theta$, where $r^2 = x^2 + y^2$, $\theta = \tan^{-1}\frac{y}{x}$.



- $2r\,r_x = 2x \Rightarrow r_x = \dfrac{x}{r} = \cos\theta;\quad r_y = \dfrac{y}{r} = \sin\theta.$
- $\theta_x = \dfrac{1}{1+(y/x)^2}\left(-\dfrac{y}{x^2}\right) = -\dfrac{y}{r^2} = -\dfrac{\sin\theta}{r};\quad \theta_y = \dfrac{x}{r^2} = \dfrac{\cos\theta}{r}.$



Derivation of the Polar Laplacian

- By the chain rule:

$$u_x = u_r r_x + u_\theta \theta_x = \cos\theta\, u_r - \frac{\sin\theta}{r} u_\theta, \qquad u_y = \sin\theta\, u_r + \frac{\cos\theta}{r} u_\theta.$$

- Differentiating again (product rule, and $\partial_x = \cos\theta\, \partial_r - \frac{\sin\theta}{r}\partial_\theta$):

$$\begin{aligned} u_{xx} &= \left(\cos\theta\, u_{rr} - \tfrac{\sin\theta}{r} u_{r\theta} + \tfrac{\sin\theta}{r^2} u_\theta\right)\cos\theta \\ &\quad - \left(\cos\theta\, u_{r\theta} - \sin\theta\, u_r - \tfrac{\sin\theta}{r} u_{\theta\theta} - \tfrac{\cos\theta}{r} u_\theta\right)\tfrac{\sin\theta}{r}, \\ u_{yy} &= \left(\sin\theta\, u_{rr} + \tfrac{\cos\theta}{r} u_{r\theta} - \tfrac{\cos\theta}{r^2} u_\theta\right)\sin\theta \\ &\quad + \left(\sin\theta\, u_{r\theta} + \cos\theta\, u_r + \tfrac{\cos\theta}{r} u_{\theta\theta} - \tfrac{\sin\theta}{r} u_\theta\right)\tfrac{\cos\theta}{r}. \end{aligned}$$

- Adding, the mixed terms and the u_θ terms cancel:

$$\boxed{u_{xx} + u_{yy} = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta}.}$$



Separation in Polar Coordinates

- To solve $u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0$, let $u(r,\theta) = R(r)\Theta(\theta)$.
- Then $u_{rr} = R''\Theta$, $u_r = R'\Theta$, $u_{\theta\theta} = R\Theta''$, and

$$R''\Theta + \frac{1}{r}R'\Theta + \frac{1}{r^2}R\Theta'' = 0 \;\Rightarrow\; (r^2R'' + rR')\Theta + R\Theta'' = 0.$$

- Hence

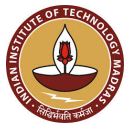
$$\frac{r^2R'' + rR'}{R} = -\frac{\Theta''}{\Theta} = \mu \quad \text{(say)}.$$

- This gives

$$r^2R'' + rR' - \mu R = 0 \quad \text{(Cauchy–Euler equation)}, \tag{2}$$

$$\Theta'' + \mu\Theta = 0. \tag{3}$$

- For (2) try $R = r^m$: $m(m-1) + m - \mu = 0$, i.e. $m^2 = \mu$.



The Three Cases

- $\mu = 0$: $r^2 R'' + rR' = 0$ gives $R_0 = A_0 \ln r + B_0$, and $\Theta_0 = C_0\theta + D_0$:

$$u_0(r,\theta) = (A_0 \ln r + B_0)(C_0\theta + D_0).$$

- $\mu = p^2 > 0$: $m = \pm p$:

$$u_p(r,\theta) = \left(A_p r^{-p} + B_p r^p\right)\left(C_p \cos p\theta + D_p \sin p\theta\right).$$

- $\mu = -p^2 < 0$: $m = \pm ip$, $r^{ip} = e^{ip \ln r}$:

$$u_p(r,\theta) = \left[A_p \cos(p \ln r) + B_p \sin(p \ln r)\right]\left(C_p e^{p\theta} + D_p e^{-p\theta}\right).$$

- Which terms survive is decided by the region: boundedness at $r = 0$, periodicity in θ, or the edge conditions.

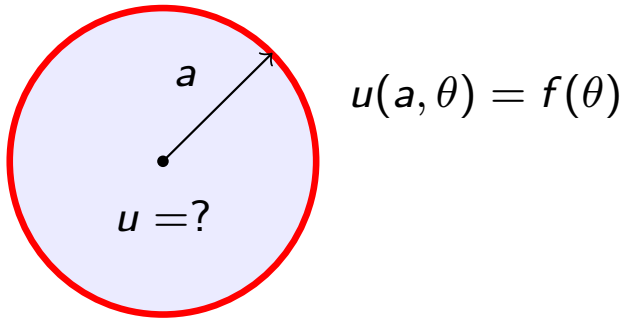


Dirichlet Problem for a Disk

- Consider

$$r^2 u_{rr} + r u_r + u_{\theta\theta} = 0, \quad 0 \le r < a, \ 0 \le \theta \le 2\pi,$$

with the boundary condition $u(a, \theta) = f(\theta)$, $0 \le \theta \le 2\pi$.



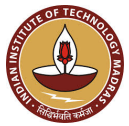
- Implicit conditions: u is bounded at $r = 0$ (the centre is inside the domain), and u is 2π-periodic in θ.

Disk: Selecting the Separated Solutions

- $\mu = 0$: $\ln r$ is not defined at $r = 0$, so $A_0 = 0$:
  $u_0 = B_0 C_0 \theta + B_0 D_0 = F_0 \theta + E_0$.
- u_0 must be periodic in θ, so $F_0 = 0$: $\quad u_0(r, \theta) = E_0$. (5)
- $\mu = -p^2 < 0$: $\cos(p \ln r)$ and $\sin(p \ln r)$ oscillate infinitely often as $r \to 0$ (and $e^{\pm p\theta}$ is not periodic), so $A_p = B_p = 0$: only the trivial solution.
- $\mu = p^2 > 0$: r^{-p} is unbounded as $r \to 0$, so $A_p = 0$:

$$u_p(r, \theta) = r^p \left(E_p \cos p\theta + F_p \sin p\theta \right), \quad E_p = B_p C_p, \; F_p = B_p D_p. \quad (6)$$

- Periodicity:
  $E_p \cos p\theta + F_p \sin p\theta = E_p \cos p(\theta + 2\pi) + F_p \sin p(\theta + 2\pi)$.



Disk: Periodicity Gives Integer p

- From the periodicity condition,

$$E_p\big[\cos p\theta - \cos(p\theta + 2p\pi)\big] + F_p\big[\sin p\theta - \sin(p\theta + 2p\pi)\big] = 0.$$

- Using sum-to-product formulas:

$$2\sin p\pi\,\big[E_p \sin(p\theta + p\pi) - F_p \cos(p\theta + p\pi)\big] = 0 \quad \text{for all } \theta.$$

- For a non-trivial solution, $\sin p\pi = 0 = \sin n\pi$, so

$$p = n, \qquad n \in \mathbb{N}. \tag{7}$$

- Together with (5), the elementary solutions are

$$u_n(r,\theta) = r^n\big(E_n \cos n\theta + F_n \sin n\theta\big), \quad n = 0, 1, 2, \ldots.$$

- By superposition,

$$u(r,\theta) = \sum_{n=0}^{\infty} r^n\big(E_n \cos n\theta + F_n \sin n\theta\big). \tag{8}$$



Disk: Fourier Coefficients

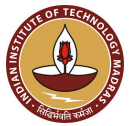
- Given $u(a,\theta) = \sum_{n=0}^{\infty} a^n \left(E_n \cos n\theta + F_n \sin n\theta\right) = f(\theta)$.
- This is the full Fourier series of f on $[0, 2\pi]$, so

$$E_0 = \frac{1}{2\pi} \int_0^{2\pi} f(\theta)\, d\theta, \tag{9}$$

$$a^n E_n = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \cos n\theta \, d\theta \;\Rightarrow\; E_n = \frac{1}{\pi a^n} \int_0^{2\pi} f(\theta) \cos n\theta \, d\theta, \quad n \geq 1 \tag{10}$$

$$a^n F_n = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \sin n\theta \, d\theta \;\Rightarrow\; F_n = \frac{1}{\pi a^n} \int_0^{2\pi} f(\theta) \sin n\theta \, d\theta, \quad n \geq 1. \tag{11}$$

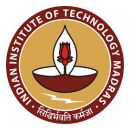
- Note $u(0,\theta) = E_0$: the value at the centre is the mean of the boundary values.
- The factor $(r/a)^n$ damps the high harmonics inside the disk.



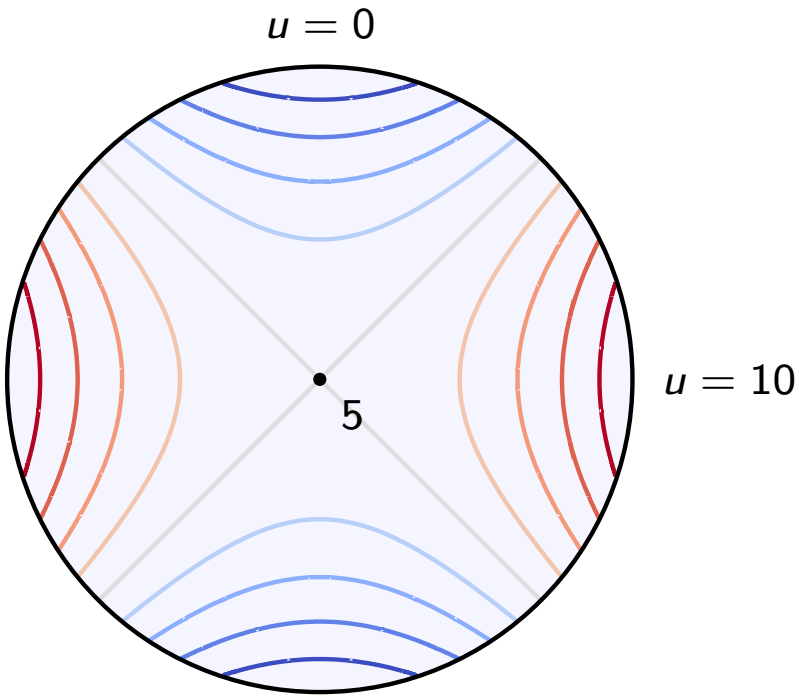
Example 5: $u(1,\theta) = 10\cos^2\theta$

- Solve $\nabla^2 u = 0$, $0 \le \theta \le 2\pi$, $0 \le r < 1$, with $u(1,\theta) = 10\cos^2\theta$.
- We know $u(r,\theta) = \sum_{n=0}^{\infty} r^n(E_n \cos n\theta + F_n \sin n\theta)$ (with $a = 1$).
- $E_0 = \dfrac{1}{2\pi}\displaystyle\int_0^{2\pi} 10\cos^2\theta\, d\theta = \dfrac{1}{2\pi}\int_0^{2\pi}(5 + 5\cos 2\theta)\, d\theta = 5.$
- $E_n = \dfrac{1}{\pi}\displaystyle\int_0^{2\pi}(5 + 5\cos 2\theta)\cos n\theta\, d\theta \Rightarrow E_2 = 5,\ E_n = 0$ for $n \neq 2$.
- $F_n = \dfrac{1}{\pi}\displaystyle\int_0^{2\pi}(5 + 5\cos 2\theta)\sin n\theta\, d\theta = 0$ for all $n \ge 1$.
- Hence

$$u(r,\theta) = 5 + 5r^2\cos 2\theta = 5 + 5(x^2 - y^2).$$



Picture: Isotherms for Example 5



- The isotherms $5 + 5(x^2 - y^2) = c$ are rectangular hyperbolas; the centre value 5 is the mean of the boundary data.
- Maximum 10 at $\theta = 0, \pi$ and minimum 0 at $\theta = \pm\frac{\pi}{2}$, both on the boundary.

Poisson's Integral Formula

- Substituting (9)–(11) in (8) and summing the geometric series gives

$$u(r,\theta) = \frac{1}{2\pi}\int_0^{2\pi} \frac{a^2 - r^2}{a^2 - 2ar\cos(\theta-\phi) + r^2} f(\phi)\, d\phi, \quad r < a.$$

- The kernel is positive and integrates to 1: u is a weighted average of the boundary values.
- Points of the boundary close to (r,θ) get larger weights.
- Mean value property: at $r = 0$,

$$u(0) = \frac{1}{2\pi}\int_0^{2\pi} f(\phi)\, d\phi.$$

- Every harmonic function equals its average over any circle centred at a point, which implies the maximum principle.



Exterior Dirichlet Problem

- Find u harmonic outside the disk, $r > a$, with $u(a, \theta) = f(\theta)$ and u bounded as $r \to \infty$.
- Now r^n is unbounded as $r \to \infty$, while r^{-n} is bounded; also $\ln r$ is excluded.
- Hence

$$u(r, \theta) = E_0 + \sum_{n=1}^{\infty} r^{-n} \left(E_n \cos n\theta + F_n \sin n\theta \right).$$

- From $u(a, \theta) = f(\theta)$: $E_0 = \frac{1}{2\pi} \int_0^{2\pi} f \, d\theta$, $E_n = \frac{a^n}{\pi} \int_0^{2\pi} f \cos n\theta \, d\theta$, $F_n = \frac{a^n}{\pi} \int_0^{2\pi} f \sin n\theta \, d\theta$.
- Example: $u(a, \theta) = \cos\theta$ gives $u = \dfrac{a}{r} \cos\theta$ outside and $u = \dfrac{r}{a} \cos\theta$ inside.
- As $r \to \infty$, $u \to E_0$, the mean of f.



Neumann Interior Problem

- Consider $r^2 u_{rr} + r u_r + u_{\theta\theta} = 0$, $0 \le r < a$, with $\dfrac{\partial u}{\partial r}(a, \theta) = f(\theta)$, $0 \le \theta \le 2\pi$.
- From (8): $\dfrac{\partial u}{\partial r} = \sum_{n=1}^{\infty} n r^{n-1} \big(E_n \cos n\theta + F_n \sin n\theta\big)$, so at $r = a$:

$$\sum_{n=1}^{\infty} n a^{n-1} \big(E_n \cos n\theta + F_n \sin n\theta\big) = f(\theta).$$

- Hence, for $n \ge 1$,

$$E_n = \frac{1}{\pi n a^{n-1}} \int_0^{2\pi} f(\theta) \cos n\theta \, d\theta, \qquad F_n = \frac{1}{\pi n a^{n-1}} \int_0^{2\pi} f(\theta) \sin n\theta \, d\theta.$$

- Compatibility: there is no constant term on the left, so we need $\int_0^{2\pi} f(\theta)\, d\theta = 0$ (zero net heat flux).
- E_0 remains arbitrary: the solution is unique only up to an additive constant.



Example 6: Neumann Problem with $f(\theta) = \cos\theta + 2\sin 3\theta$

- Solve $\Delta u = 0$ for $r < a$ with $u_r(a,\theta) = \cos\theta + 2\sin 3\theta$.
- Compatibility: $\int_0^{2\pi} (\cos\theta + 2\sin 3\theta)\, d\theta = 0$. ✓.
- Comparing coefficients in $\sum n a^{n-1}(E_n \cos n\theta + F_n \sin n\theta) = \cos\theta + 2\sin 3\theta$:
- $1 \cdot a^0 E_1 = 1 \Rightarrow E_1 = 1, \quad 3a^2 F_3 = 2 \Rightarrow F_3 = \dfrac{2}{3a^2}$, all others zero.
- Hence

$$u(r,\theta) = E_0 + r\cos\theta + \frac{2r^3}{3a^2}\sin 3\theta, \quad E_0 \text{ arbitrary.}$$

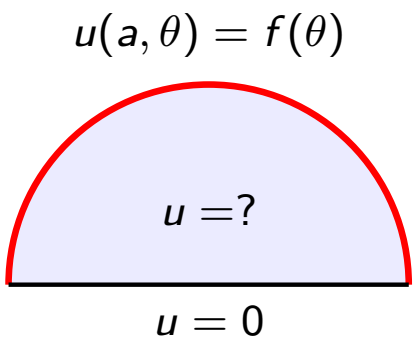
- With $f = 1$ instead, there would be no solution: heat would flow in everywhere and never reach a steady state.



Semicircular Domain

- Consider $r^2 u_{rr} + r u_r + u_{\theta\theta} = 0$, $0 \le \theta \le \pi$, $0 \le r < a$, with

$$u(a, \theta) = f(\theta), \ 0 \le \theta \le \pi; \qquad u(r, 0) = u(r, \pi) = 0.$$



- Let $u = R(r)\Theta(\theta)$: $r^2 R'' + rR' - \mu R = 0$ (1), $\Theta'' + \mu\Theta = 0$ (2).
- $u(r, 0) = R(r)\Theta(0) = 0$ and $u(r, \pi) = R(r)\Theta(\pi) = 0$ give $\Theta(0) = 0 = \Theta(\pi)$. (3)



Semicircle: Solution

- For $\mu = 0$ and $\mu = -p^2$: $\Theta \equiv 0$ (homework).
- For $\mu = p^2$: $\Theta_p = C_p \cos p\theta + D_p \sin p\theta$; $\Theta_p(0) = 0 \Rightarrow C_p = 0$.
- $\Theta_p(\pi) = D_p \sin p\pi = 0 \Rightarrow p = n$, $n \in \mathbb{N}$; so $\Theta_n = D_n \sin n\theta$.
- For $\mu = n^2$: $r^2 R'' + rR' - n^2 R = 0$, $R = r^m$, $m^2 - n^2 = 0$, so $R_n = A_n r^n + B_n r^{-n}$.
- $u_n = \left(E_n r^n + F_n r^{-n}\right) \sin n\theta$. Since u_n is bounded as $r \to 0$, $F_n = 0$ for all n.
- By superposition,

$$u(r,\theta) = \sum_{n=1}^{\infty} E_n r^n \sin n\theta, \qquad E_n = \frac{2}{\pi a^n} \int_0^{\pi} f(\theta) \sin n\theta \, d\theta,$$

since $u(a,\theta) = \sum E_n a^n \sin n\theta = f(\theta)$ is a Fourier sine series on $[0,\pi]$.



Example 7: $u(1,\theta) = 10\,\theta(\pi - \theta)$

- Solve $\Delta u = 0$, $0 \le \theta \le \pi$, $0 \le r < 1$, with $u(1,\theta) = 10\theta(\pi - \theta)$, $u(r,0) = u(r,\pi) = 0$.
- $u(r,\theta) = \sum_{n=1}^{\infty} E_n r^n \sin n\theta$, $\quad E_n = \dfrac{2}{\pi}\displaystyle\int_0^{\pi} 10\,\theta(\pi - \theta)\sin n\theta\, d\theta$.
- Integrating by parts twice:

$$\int_0^{\pi} \theta(\pi - \theta)\sin n\theta\, d\theta = \frac{2}{n^3}\left(1 - (-1)^n\right).$$

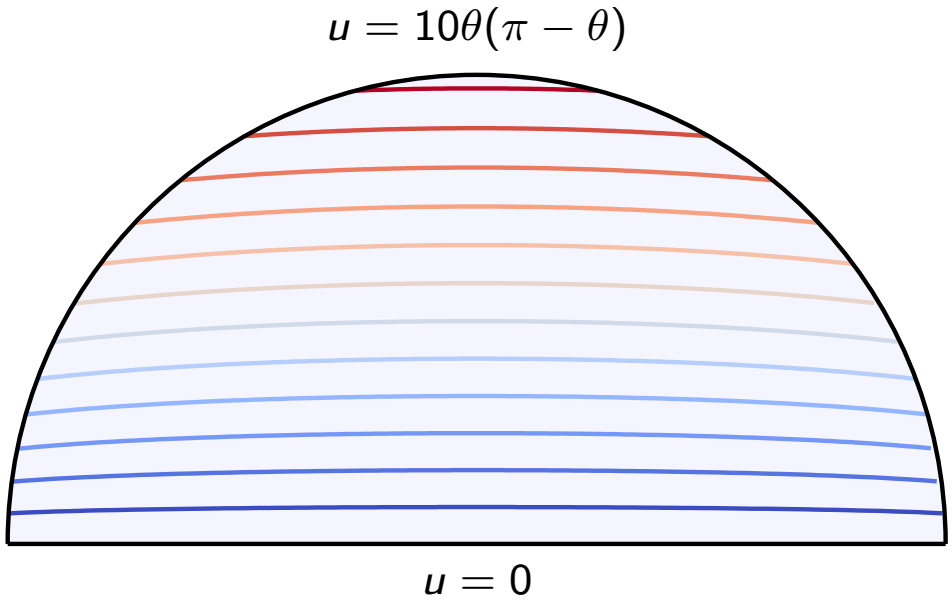
- Hence

$$E_n = \frac{40}{\pi n^3}\left(1 - (-1)^n\right) = \begin{cases} \dfrac{80}{\pi n^3}, & n \text{ odd}, \\ 0, & n \text{ even}, \end{cases}$$

$$u(r,\theta) = \frac{80}{\pi}\sum_{n=1}^{\infty} \frac{r^{2n-1}}{(2n-1)^3}\sin(2n-1)\theta.$$



Picture: Isotherms for Example 7



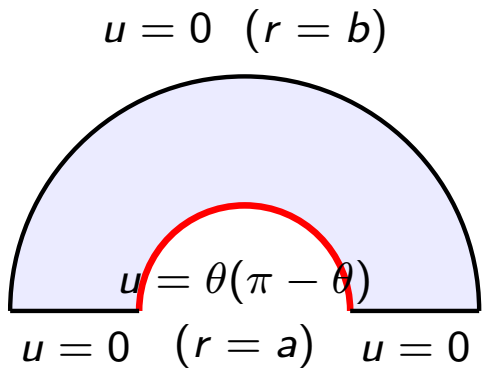
- Levels $u = 2, 4, \ldots, 24$; the hottest boundary point is $\theta = \pi/2$, where $u = 10\pi^2/4 \approx 24.7$.
- All isotherms end at the corners $(\pm 1, 0)$, where the data drop to zero.



Semicircular Annulus

- Solve $r^2 u_{rr} + r u_r + u_{\theta\theta} = 0$, $a \le r \le b$, $0 \le \theta \le \pi$, with

  $u(r,0) = 0 = u(r,\pi), \quad a \le r \le b; \qquad u(a,\theta) = \theta(\pi - \theta), \quad u(b,\theta) = 0.$



- As in the semicircle, $u_n(r,\theta) = \left(E_n r^n + F_n r^{-n}\right) \sin n\theta$; now $r = 0$ is not in the region, so F_n need not vanish.



Semicircular Annulus: Solution

- $u(b,\theta) = 0 \Rightarrow E_n b^n + F_n b^{-n} = 0 \Rightarrow F_n = -b^{2n} E_n$, so

$$u_n(r,\theta) = E_n\left(r^n - b^{2n} r^{-n}\right) \sin n\theta = E_n \, \frac{r^{2n} - b^{2n}}{r^n} \, \sin n\theta.$$

- By superposition $u = \sum_{n=1}^{\infty} E_n \dfrac{r^{2n} - b^{2n}}{r^n} \sin n\theta$, and $u(a,\theta) = \theta(\pi - \theta)$ gives

$$E_n \, \frac{a^{2n} - b^{2n}}{a^n} = \frac{2}{\pi} \int_0^{\pi} \theta(\pi - \theta) \sin n\theta \, d\theta = -\frac{4}{\pi n^3}\left[(-1)^n - 1\right].$$

- So $E_n = \dfrac{4a^n}{(b^{2n} - a^{2n})\pi n^3}\left[(-1)^n - 1\right]$, and the desired solution is

$$u(r,\theta) = \frac{8}{\pi} \sum_{n \text{ odd}} \left(\frac{b^{2n} - r^{2n}}{b^{2n} - a^{2n}}\right) \frac{a^n}{r^n} \, \frac{\sin n\theta}{n^3}.$$

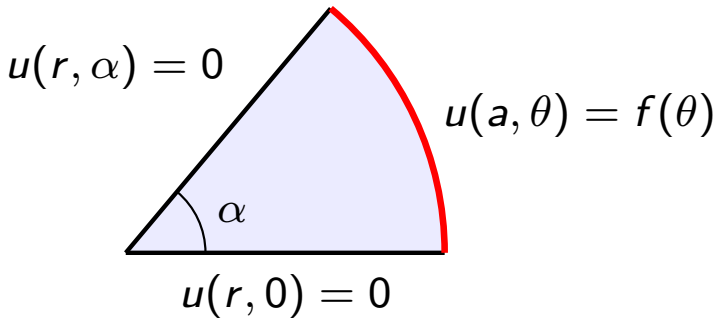
- Check: at $r = b$ the bracket vanishes; at $r = a$ it equals 1.



Wedge P1: Dirichlet Conditions

- Solve $r^2 u_{rr} + r u_r + u_{\theta\theta} = 0$, $0 < r < a$, $0 < \theta < \alpha$, with

$$u(r,0) = 0, \quad u(r,\alpha) = 0, \quad u \text{ bounded as } r \to 0, \quad u(a,\theta) = f(\theta).$$



- With $u = R\Theta$: $\Theta(0) = \Theta(\alpha) = 0$. For $\mu = 0$, $\mu = -p^2$: $\Theta \equiv 0$.
- For $\mu = p^2$: $\Theta = A\cos p\theta + B \sin p\theta$; $A = 0$ and $\sin p\alpha = 0 \Rightarrow p = \dfrac{n\pi}{\alpha}$, $\Theta_n = \sin \dfrac{n\pi\theta}{\alpha}$.



Wedge P1: Solution

- Solve (1) with $\mu = p^2 = \dfrac{n^2\pi^2}{\alpha^2}$: $R_n(r) = E_n r^{n\pi/\alpha} + F_n r^{-n\pi/\alpha}$.
- So $u_n(r,\theta) = \sin\dfrac{n\pi\theta}{\alpha}\left(E_n r^{n\pi/\alpha} + F_n r^{-n\pi/\alpha}\right)$.
- Since u_n is bounded as $r \to 0$, $F_n = 0$ for all n:

$$u(r,\theta) = \sum_{n=1}^{\infty} E_n r^{n\pi/\alpha} \sin\frac{n\pi\theta}{\alpha}.$$

- From $u(a,\theta) = f(\theta) = \sum E_n a^{n\pi/\alpha} \sin\frac{n\pi\theta}{\alpha}$:

$$E_n = \frac{2}{\alpha\, a^{n\pi/\alpha}} \int_0^{\alpha} f(\theta) \sin\frac{n\pi\theta}{\alpha}\, d\theta.$$

- For $\alpha = \pi$ this reduces to the semicircle; the exponents $n\pi/\alpha$ need not be integers.



Wedge P2: Non-homogeneous Edge Values

- Solve $\Delta u = 0$, $0 < r < a$, $0 < \theta < \alpha$, with $u(r,0) = u_0$, $u(r,\alpha) = u_1$, u bounded as $r \to 0$, $u(a,\theta) = f(\theta)$.
- The simplest function of θ only satisfying the edge conditions is

$$w(\theta) = (u_1 - u_0)\frac{\theta}{\alpha} + u_0, \qquad w_{\theta\theta} = 0,\ w(0) = u_0,\ w(\alpha) = u_1.$$

- w is harmonic: $\Delta w = \frac{1}{r^2} w_{\theta\theta} = 0$.
- Let $u(r,\theta) = w(\theta) + v(r,\theta)$. Then $v_{rr} + \frac{1}{r} v_r + \frac{1}{r^2} v_{\theta\theta} = 0$ and

$$v(r,0) = u_0 - u_0 = 0, \quad v(r,\alpha) = u_1 - u_1 = 0, \quad v(a,\theta) = f(\theta) - w(\theta).$$

- This is problem P1 for v, with data $f - w$.



Wedge P2: Solution

- By P1,

$$v(r,\theta) = \sum_{n=1}^{\infty} E_n r^{n\pi/\alpha} \sin\frac{n\pi\theta}{\alpha}, \quad E_n = \frac{2}{\alpha\, a^{n\pi/\alpha}} \int_0^{\alpha} \big(f(\theta) - w(\theta)\big) \sin\frac{n\pi\theta}{\alpha}$$

- Therefore

$$u(r,\theta) = (u_1 - u_0)\frac{\theta}{\alpha} + u_0 + \sum_{n=1}^{\infty} E_n r^{n\pi/\alpha} \sin\frac{n\pi\theta}{\alpha}.$$

- The term $w(\theta)$ plays the role of the steady state u_s in the heat equation (Case IV).
- Geometrically: w is constant along each ray $\theta =$ const and changes linearly with the angle.



Wedge P3: Insulated Edges

- Solve $\Delta u = 0$, $0 < \theta < \alpha$, $0 < r < a$, with $u_\theta(r, 0) = 0$, $u_\theta(r, \alpha) = 0$, $u(a, \theta) = f(\theta)$.
- With $u = R\Theta$: $r^2 R'' + rR' - \mu R = 0$, $\Theta'' + \mu\Theta = 0$, $\Theta'(0) = \Theta'(\alpha) = 0$.
- $\mu = 0$: $\Theta_0 = B_0$, $R_0 = c_0 + d_0 \ln r$; boundedness gives $d_0 = 0$: $u_0 = E_0$.
- $\mu = -p^2$: $\Theta \equiv 0$. $\mu = p^2$: $\Theta = C\cos p\theta + D\sin p\theta$; $\Theta'(0) = 0 \Rightarrow D = 0$; $\Theta'(\alpha) = 0 \Rightarrow p = \frac{n\pi}{\alpha}$.
- $\Theta_n = \cos\frac{n\pi\theta}{\alpha}$, $R_n = E_n r^{n\pi/\alpha} + F_n r^{-n\pi/\alpha}$, and boundedness gives $F_n = 0$:

$$u(r, \theta) = E_0 + \sum_{n=1}^{\infty} E_n r^{n\pi/\alpha} \cos\frac{n\pi\theta}{\alpha},$$

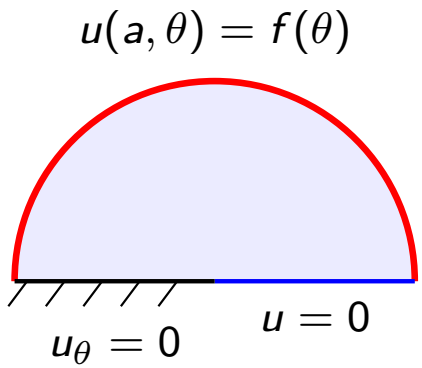
$$E_0 = \frac{1}{\alpha}\int_0^\alpha f(\theta)\, d\theta, \qquad E_n = \frac{2}{\alpha\, a^{n\pi/\alpha}}\int_0^\alpha f(\theta)\cos\frac{n\pi\theta}{\alpha}\, d\theta.$$



Wedge P4: Mixed Conditions (Crack-like Problem)

- Solve $\Delta u = 0$ in the half disk $0 < r < a$, $0 < \theta < \pi$, with

$$u(r,0) = 0, \qquad \frac{\partial u}{\partial \theta}(r,\pi) = 0, \qquad u(a,\theta) = f(\theta).$$



- $\Theta(0) = 0$, $\Theta'(\pi) = 0$. For $\mu = 0$ and $\mu = -p^2$: $\Theta \equiv 0$.
- For $\mu = p^2$: $\Theta = A\cos p\theta + B\sin p\theta$; $A = 0$;
  $\Theta'(\pi) = Bp\cos p\pi = 0 \Rightarrow p = \dfrac{2n-1}{2}$, $n \in \mathbb{N}$.

Wedge P4: Solution

- So $\Theta_n(\theta) = B_n \sin \dfrac{(2n-1)\theta}{2}$ and $R_n(r) = C_n r^{(2n-1)/2} + D_n r^{-(2n-1)/2}$.
- $u_n = \left(E_n r^{(2n-1)/2} + F_n r^{-(2n-1)/2}\right) \sin \frac{(2n-1)\theta}{2}$; boundedness as $r \to 0$ gives $F_n = 0$.
- By superposition,

$$u(r,\theta) = \sum_{n=1}^{\infty} E_n r^{(2n-1)/2} \sin \frac{(2n-1)\theta}{2},$$

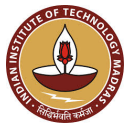
$$E_n = \frac{2}{\pi a^{(2n-1)/2}} \int_0^{\pi} f(\theta) \sin \frac{(2n-1)\theta}{2}\, d\theta.$$

- The leading term behaves like $\sqrt{r}$: $u_r \sim r^{-1/2}$ is unbounded at the tip $r = 0$.
- This square-root singularity at the point where the condition changes is typical of cracks in elasticity.



Summary

- The variable with two homogeneous boundary conditions carries the eigenvalue problem (sines/cosines).
- The other variable gets sinh/cosh (rectangle), decaying exponentials (strip), or powers $r^{\pm p}$ (polar).
- Rectangle: $\sin \frac{n\pi x}{a} \sinh \frac{n\pi(b-y)}{a}$ etc.; several hot edges by superposition.
- Strip: $\sin \frac{n\pi x}{L} e^{-n\pi y/L}$; insulated sides keep a constant mode.
- Disk: $r^n(E_n \cos n\theta + F_n \sin n\theta)$; outside: r^{-n}; centre value = mean of boundary values.
- Semicircle: $r^n \sin n\theta$; annulus: $E_n(r^{2n} - b^{2n})r^{-n} \sin n\theta$.
- Wedge of angle α: $r^{n\pi/\alpha} \sin \frac{n\pi\theta}{\alpha}$ (or cos); mixed edges: $r^{(2n-1)/2}$.
- Neumann problems need $\oint \partial u/\partial n = 0$ and are unique up to a constant.



Homework: Polar Regions

- (1) Solve $\Delta u = 0$ in $r < 1$ with $u(1, \theta) = \sin^3 \theta$. (Use $\sin^3 \theta = \frac{3}{4} \sin \theta - \frac{1}{4} \sin 3\theta$.).
- (2) Solve $\Delta u = 0$ in $r < a$ with $u(a, \theta) = 1$ for $0 < \theta < \pi$ and 0 for $\pi < \theta < 2\pi$.
- (3) Solve $\Delta u = 0$ for $r > 1$ with $u(1, \theta) = 1 + \cos 2\theta$, u bounded at infinity.
- (4) Solve $\Delta u = 0$ in the annulus $1 < r < 2$ with $u(1, \theta) = 0$, $u(2, \theta) = \sin \theta$.
- (5) Solve $\Delta u = 0$ in the semicircle $r < 1$, $0 < \theta < \pi$, with $u(1, \theta) = \sin 2\theta$, $u(r, 0) = u(r, \pi) = 0$.
- (6) Solve the wedge problem P1 with $\alpha = \pi/2$, $a = 1$, $f(\theta) = \theta(\pi/2 - \theta)$.
- (7) For which f does the Neumann problem $u_r(1, \theta) = f(\theta)$ in the unit disk have a solution? Solve it for $f = \sin 2\theta$.

